

Education

- 2024 – Now **Hong Kong Polytechnic University**, Ph.D. in Computing, GPA: 4.1/4.3.
Advisor: Prof. **Wenqi Fan** and Prof. **Qing Li**
- 2021 – 2024 **Sichuan University**, M.E. in Computer Science and Technology, GPA: 3.76/4.
Advisor: Prof. **Min Zhu**, Lab: Vision Computing Lab
- 2017 – 2021 **Sichuan University**, B.E. in Computer Science and Technology, GPA: 3.77/4.
Under the Wu Yuzhang Honors Program

Research Interests

AI for Science, Protein Language Models, Large Language Models, Diffusion Models

Publications & Preprints (First Author)

- 2026 **Yi Zhou**, Qipeng Wang, Yunqing Liu, Jun Xia, Qing Li, Wenqi Fan. "Unlocking Multimodal Protein Language Models at Inference Time". *Conference on Empirical Methods in Natural Language Processing (EMNLP Main)*, 2026. (CCF-B)
- Systematically studied the inference design space of multimodal protein language models by exploring three bottom-up strategies (vanilla sampling, classifier-free guidance, and reward-guided beam search), achieving substantial parameter-free performance gains and revealing base model behaviors.
- Yi Zhou**, Haohao Qu, Yunqing Liu, Shanru Lin, Le Song, Wenqi Fan. "HD-Prot: A Protein Language Model for Joint Sequence-Structure Modeling with Continuous Structure Tokens". *International Conference on Knowledge Discovery and Data Mining (KDD)*, 2026. (CCF-A)
- Proposed HD-Prot, a *Hybrid Diffusion Protein* language model that integrates continuous structure tokens into a discrete protein language model via a unified diffusion process, achieving competitive and efficient protein sequence-structure joint modeling.
- 2025 **Yi Zhou**, Wenqi Fan, Shijie Wang, Yuyao Yan, Hui Liu, Qian Zhao, Le Song, Qing Li. "Computational Protein Science in the Era of Large Language Models (LLMs)". *ACM Computing Surveys (CSUR)*, Under Review.
- Authored a comprehensive review on protein language models, systematically summarizing their foundational sequence-structure-function knowledge modeling and discussing their adaptations, applications and future directions in protein structure prediction, function prediction, and protein design.
- 2024 **Yi Zhou**, Xian Guan, Meixuan Wu, Chengzhou Ouyang, Min Zhu. "Timely-MDA: A Benchmark for Generalizable MiRNA-Disease Association Prediction". *International Conference on Bioinformatics and Biomedicine (BIBM)*, 2024. (CCF-B)
- 2022 **Yi Zhou**, Xinyi Wang, Lin Yao, Min Zhu. "LDAformer: Predicting LncRNA-Disease Associations based on Topological Feature Extraction and Transformer Encoder". *Briefings in Bioinformatics (BIB)*, 2022. (JCR-Q1, IF: 13.994, CCF-B)

Publications & Preprints (Collaboration)

- 2026 Yunqing Liu, **Yi Zhou**, Wenqi Fan. "Enhancing Molecular Property Predictions by Learning from Bond Modelling and Interactions". *International Conference on Learning Representations (ICLR)*, 2026. (CCF-A)

Yunqing Liu, **Yi Zhou**, Jun Xia, Wenqi Fan, Qing Li. "Geometric Flow Matching for Conformation Generation via Manifold Decomposition". *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 2026, Under Review.

Tianzhong Lan, Weili Jiang, Yisong Liu, **Yi Zhou**, Junqi Bai, Si Yong Yeo, Xulei Yang, Min Zhu. "Wavelet-Driven Cross-Domain Consistency for Mixed-Supervised 3D Tumor Segmentation". *European Conference on Computer Vision (ECCV)*, 2026. (CCF-B)

2025 Lin Gan, Xinyi Wang, **Yi Zhou**, Min Zhu. "Protein-binding RNA Prediction Based on Integrated Sequence-Structure-Function Pre-training". *IEEE Transactions on Computational Biology and Bioinformatics (TCBB)*, 2025. (JCR-Q1)

Meixuan Wu, Yizhou Yang, **Yi Zhou**, Kai Wang, Junqi Bai, Min Zhu. "PLHGMDA: Pre-trained Language Model and Heterogeneous Graph Neural Network for MiRNA-Disease Association Prediction". *International Conference on Intelligent Computing (ICIC)*, 2025. (CCF-C, Oral)

2024 Wenwen Gao, Shangsong Liu, **Yi Zhou**, Fengjie Wang, Feng Zhou, Min Zhu. "GBDT4CTRVis: Visual Analytics of Gradient Boosting Decision Tree for Advertisement Click-Through Rate Prediction". *Journal of Visualization (JoV)*, 2024.

Jiamin Zhu, Meixuan Wu, **Yi Zhou**, Nan Cao, Haotian Zhu, Min Zhu. "Dowsing: A Task-Driven Approach for Multiple-View Visualizations Dynamic Recommendation". *Journal of Visualization (JoV)*, 2024.

Tutorials

2026 Yujuan Ding, Liangbo Ning, Ziran Liang, Chun-Hin Chan, **Yi Zhou**, Haohao Qu, Wenqi Fan. "Towards Autonomous Web Interaction: Empowering WebAgents by Large Foundation Models". *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*, 2026. (CCF-C)

Patents

2024 Min Zhu, Meixuan Wu, **Yi Zhou**. "A MicroRNA-Disease Association Prediction Method Based on Graph Neural Networks". CN119763668B.

2022 Min Zhu, **Yi Zhou**, Xinyi Wang, Lin Yao. "Method and System for Long Non-coding RNA-Disease Association Prediction Based on Self-Attention Mechanism". CN115171780B.

Min Zhu, Xiyao Li, Chunlin Long, **Yi Zhou**, Xinyi Wang. "A Prediction Method for Chromatin Topological Association Domain Boundary Based on Multimodal Fusion". CN115831217B.

Min Zhu, Jiamin Zhu, Meixuan Wu, **Yi Zhou**, Haotian Zhu. "A Dynamic Visualization Recommendation Method Based on User Tasks". CN116204704B.

Min Zhu, Meixuan Wu, Jiamin Zhu, **Yi Zhou**, Haotian Zhu. "An Analytical Task Perception Method that Integrates Deep Learning Models and Rules". CN116303737B.

2021 Min Zhu, Fuqiu Chen, Chunlin Long, **Yi Zhou**, Xinyi Wang. "A Visualization Method for Chromatin Hierarchy Analysis Based on Genetic Data". CN113946730B.

Teaching & Services

2023 - 2024 **Teaching Assistant**, Sichuan University.

- Data Visualization (311301030) × 2

2024 - Now **Teaching Assistant**, Hong Kong Polytechnic University.

- Introduction to Artificial Intelligence and Data Analytic (COMP1004) × 3

- Computational Thinking and Problem Solving (COMP1010)

Reviewer.

- Conference: KDD AI for Sciences Track - 2026/2027

- Journal: Expert Systems With Applications (ESWA)

Honors & Awards

- 2024 PolyU Research Postgraduate Scholarship (PRPgS)
Outstanding Graduate of Sichuan University
Outstanding Undergraduate Teaching Assistant at Sichuan University
- 2023 Outstanding Graduate Student of Sichuan University
First-class of Excellent Graduate Scholarship by Sichuan University
Tencent Scholarship of Sichuan University
- 2021 Certificate of Honor from Wu Yuzhang Honors College

Skills

Programming Python, PyTorch, SQL, Linux commands, Git, LaTeX
Languages Mandarin, English (IELTS: 7.0)